



De Moivre's Theorem

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Learning Objectives

- Convert complex numbers from rectangular form to trigonometric (polar) form
- Apply De Moivre's Theorem to raise complex numbers to integer powers
- Express the final result in rectangular form $a + bi$

Use De Moivre's Theorem to evaluate each power of a complex number and express your final answer in rectangular form.

1. Evaluate the following using De Moivre's Theorem.

$$(-1 + i\sqrt{3})^{12}$$

Answer: _____

2. Use De Moivre's Theorem to evaluate.

$$(1 + i)^5$$

Answer: _____

3. Evaluate using De Moivre's Theorem.

$$(1 - i)^8$$

Answer: _____

4. Evaluate using De Moivre's Theorem.

$$(\sqrt{3} + i)^6$$

Answer: _____

5. Use De Moivre's Theorem to simplify.

$$[2(\cos 40^\circ + i\sin 40^\circ)]^3$$

Answer: _____

6. Evaluate using De Moivre's Theorem.

$$(-1 - i)^4$$

Answer: _____

7. Use De Moivre's Theorem to evaluate.

$$(2 + 2i)^6$$

Answer: _____



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8. Evaluate the power using De Moivre's Theorem.

$$[3(\cos 60^\circ + i\sin 60^\circ)]^4$$

Answer: _____

9. Evaluate using De Moivre's Theorem.

$$(\sqrt{3} - i)^5$$

Answer: _____

10. Use De Moivre's Theorem to evaluate.

$$(1 + i\sqrt{3})^{10}$$

Answer: _____





Remind students to first convert to trigonometric form $z = r(\cos \theta + i \sin \theta)$ before applying $z^n = r^n(\cos n\theta + i \sin n\theta)$.

Solutions

1. Evaluate the following using De Moivre's Theorem.

$$(-1 + i\sqrt{3})^{12}$$

→ Find $r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2$.

→ Find θ : the point $(-1, \sqrt{3})$ lies in quadrant II, so $\theta = 120^\circ$.

→ Write in trig form: $2(\cos 120^\circ + i \sin 120^\circ)$.

→ Apply De Moivre: $2^{12}(\cos(12 \cdot 120^\circ) + i \sin(12 \cdot 120^\circ)) = 4096(\cos 1440^\circ + i \sin 1440^\circ)$.

→ Reduce $1440^\circ \bmod 360^\circ = 0^\circ$, so the value is $4096(1 + 0i) = 4096$.

Answer: **4096**

2. Use De Moivre's Theorem to evaluate.

$$(1 + i)^5$$

→ Find $r = \sqrt{1^2 + 1^2} = \sqrt{2}$.

→ Find $\theta = 45^\circ$ since the point $(1, 1)$ is in quadrant I.

→ Write in trig form: $\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$.

→ Apply De Moivre: $(\sqrt{2})^5(\cos 225^\circ + i \sin 225^\circ) = 4\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$.

→ Substitute $\cos 225^\circ = -\sqrt{2}/2$ and $\sin 225^\circ = -\sqrt{2}/2$ to get $4\sqrt{2} \cdot (-\sqrt{2}/2 - i\sqrt{2}/2) = -4 - 4i$.

Answer: **-4 - 4i**

3. Evaluate using De Moivre's Theorem.

$$(1 - i)^8$$

→ Compute $r = \sqrt{1^2 + 1^2} = \sqrt{2}$.

→ Find $\theta = -45^\circ$ (or 315°) since $(1, -1)$ is in quadrant IV.

→ Trig form: $\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ))$.

→ Apply De Moivre: $(\sqrt{2})^8(\cos(-360^\circ) + i \sin(-360^\circ)) = 16(1 + 0i)$.

→ Final answer is 16.

Answer: **16**

4. Evaluate using De Moivre's Theorem.

$$(\sqrt{3} + i)^6$$

→ Compute $r = \sqrt{3 + 1} = 2$.

→ Find $\theta = 30^\circ$ since the point $(\sqrt{3}, 1)$ is in quadrant I.

→ Trig form: $2(\cos 30^\circ + i \sin 30^\circ)$.

→ Apply De Moivre: $2^6(\cos 180^\circ + i \sin 180^\circ) = 64(-1 + 0i)$.

→ Final answer: -64.

Answer: **-64**



5. Use De Moivre's Theorem to simplify.

$$[2(\cos 40^\circ + i \sin 40^\circ)]^3$$

- Raise the modulus to the power: $2^3 = 8$.
- Multiply the angle by the exponent: $3 \times 40^\circ = 120^\circ$.
- Write: $8(\cos 120^\circ + i \sin 120^\circ)$.
- Substitute $\cos 120^\circ = -1/2$ and $\sin 120^\circ = \sqrt{3}/2$.
- Multiply: $8(-1/2) + 8i(\sqrt{3}/2) = -4 + 4i\sqrt{3}$.

Answer: $8(\cos 120^\circ + i \sin 120^\circ) = -4 + 4i\sqrt{3}$

6. Evaluate using De Moivre's Theorem.

$$(-1 - i)^4$$

- Compute $r = \sqrt{1 + 1} = \sqrt{2}$.
- The point $(-1, -1)$ is in quadrant III, so $\theta = 225^\circ$.
- Trig form: $\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$.
- Apply De Moivre: $(\sqrt{2})^4 (\cos 900^\circ + i \sin 900^\circ) = 4(\cos 180^\circ + i \sin 180^\circ)$.
- Substitute: $4(-1 + 0i) = -4$.

Answer: -4

7. Use De Moivre's Theorem to evaluate.

$$(2 + 2i)^6$$

- Compute $r = \sqrt{4 + 4} = 2\sqrt{2}$.
- Find $\theta = 45^\circ$ since the point $(2, 2)$ is in quadrant I.
- Trig form: $2\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$.
- Apply De Moivre: $(2\sqrt{2})^6 (\cos 270^\circ + i \sin 270^\circ) = 512(0 - i) = -512i$.

Answer: $-512i$

8. Evaluate the power using De Moivre's Theorem.

$$[3(\cos 60^\circ + i \sin 60^\circ)]^4$$

- Raise the modulus: $3^4 = 81$.
- Multiply the angle: $4 \times 60^\circ = 240^\circ$.
- Write: $81(\cos 240^\circ + i \sin 240^\circ)$.
- Substitute $\cos 240^\circ = -1/2$ and $\sin 240^\circ = -\sqrt{3}/2$.
- Multiply: $81(-1/2) + 81i(-\sqrt{3}/2) = -81/2 - (81\sqrt{3}/2)i$.

Answer: $-\frac{81}{2} - \frac{81\sqrt{3}}{2}i$

9. Evaluate using De Moivre's Theorem.

$$(\sqrt{3} - i)^5$$

- Compute $r = \sqrt{3 + 1} = 2$.
- The point $(\sqrt{3}, -1)$ is in quadrant IV, so $\theta = -30^\circ$.
- Trig form: $2(\cos(-30^\circ) + i \sin(-30^\circ))$.
- Apply De Moivre: $2^5 (\cos(-150^\circ) + i \sin(-150^\circ)) = 32(-\sqrt{3}/2 - i/2)$.
- Multiply: $-16\sqrt{3} - 16i$.

Answer: $-16\sqrt{3} - 16i$



10. Use De Moivre's Theorem to evaluate.

$$(1 + i\sqrt{3})^{10}$$

→ Compute $r = \sqrt{1 + 3} = 2$.

→ Find $\theta = 60^\circ$ since the point $(1, \sqrt{3})$ is in quadrant I.

→ Trig form: $2(\cos 60^\circ + i \sin 60^\circ)$.

→ Apply De Moivre: $2^{10}(\cos 600^\circ + i \sin 600^\circ) = 1024(\cos 240^\circ + i \sin 240^\circ)$.

→ Substitute $\cos 240^\circ = -1/2$ and $\sin 240^\circ = -\sqrt{3}/2$: $1024(-1/2 - i\sqrt{3}/2) = -512 - 512i\sqrt{3}$.

Answer: $-512 - 512i\sqrt{3}$

